

Question	Answer	Marks
1(a)(i)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(a)(ii)	(near Earth's surface change in) height $\ll$ radius or height <u>much</u> less than radius	B1
	potential inversely proportional to radius and radius approximately constant (so potential approximately constant)	B1
1(b)	initial kinetic energy = (–) potential energy (at surface) or $\frac{1}{2}mv^2 = GMm/r$	B1
	$v^2 = (2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / (0.5 \times 3.5 \times 10^6)$	C1
	$v = 2.4 \times 10^3 \text{ m s}^{-1}$	A1

Question	Answer	Marks
2(a)	sum of potential and kinetic energies (of molecules/atoms/particles)	B1